

LESSON 118 (0.5 GROUND and 1.5 DUAL)

Lesson Objective

The purpose of this lesson is for the learner to keep practicing emergency procedures and ground reference maneuvers. During this lesson the learner will perform the procedure for an engine failure in flight with emphasis on the emergency approach and landing pattern. The learner should be able to make radio calls with little instructor guidance.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Warrior Maneuver's Manual.
 - Emergency checklists.
 - V speed and Emergency quiz review.
- Student questions.

References

- Advisory Circular. ([Search Tool](#))
 - Non-Towered Airport Flight Operations. ([AC 90-66](#))
- Aeronautical Information Manual. ([AIM](#))
 - Chapter 06 - Emergency Procedures.
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))

Instructor Guidance

- Begin the briefing by quizzing the student on the emergency checklists and V-speeds from their homework. With Lesson 121 pre-solo tests approaching, these must be memorized — not just reviewed. If the student cannot recite the engine failure ABCD flow and key V-speeds accurately, address it before the flight and make clear that this knowledge is a prerequisite for solo.
- This lesson is a proficiency-building lesson — the student has already been introduced to emergency procedures and ground reference maneuvers. The expectation this time is meaningful improvement. Use the student's performance from Lesson 117 to identify specific areas to focus on and brief those areas explicitly before departing.
- For emergency procedures, the student should now be performing independently ("student says, student does") without demonstration from you first. If you find yourself needing to re-demonstrate, treat it as a signal to assess whether the student is ready to progress or whether additional practice is needed before moving forward.
- During the engine failure in flight, resist the urge to prompt the student through the ABCD flow. Allow them to work through it and note any hesitation or missed steps — this is a more accurate representation of their real-world readiness than a coached performance. Debrief

specifically on any gaps.

- Wire strike avoidance must be addressed every single time an engine failure is simulated. Florida's low terrain and high density of power lines make this a genuine operational hazard — it is not a box-checking exercise.
- For the emergency descent, closely monitor airspeed throughout. The most common error at this stage is allowing airspeed to build beyond VFE before or after flap extension. Guard against it and correct it immediately.
- Ground reference maneuvers in this lesson should be performed with minimal instructor guidance per the completion standard. Before entering each maneuver, require the student to independently assess wind direction using available cues. If they cannot determine the wind, prompt them to do so rather than providing the answer — wind assessment is a skill that must be owned by the student before solo.
- The traffic pattern quiz during the ground portion should be thorough — entry procedures, airspeeds, flap settings, and power settings for each leg. A student who is approaching solo flight must be able to recite and execute a complete pattern without coaching.
- Radio calls should require little to no input from you at this stage. If the student is still hesitating on calls or using incorrect phraseology, address it directly and set a clear expectation for improvement before Lesson 121.
- During the post-brief, be honest and specific with your critique. The student is one lesson away from Lesson 119 which further develops pattern and stall proficiency, and two lessons away from the pre-solo tests in Lesson 121. If there are meaningful gaps in emergency procedure recall, landing consistency, or ground reference maneuver performance, the student needs to know now so they can prepare accordingly.

Lesson Content (Ground)

Engine Failure in Flight (Emphasis on Pattern)

- Order of operations
 - A - Airspeed
 - Airspeed comes first and is critical to maintain throughout the maneuver because it guarantees you the most possible time in the air to make a safe landing.
 - Review why we use V_{Glide} (73) and not another airspeed.
 - Total drag curve.
 - Show where 73 is and why going at 63 or 83 (for example) would be worse.
 - B - Best field
 - After you can keep yourself in the air for as long as possible, the next most important consideration is where to put the airplane when that time runs out.
 - C - Checklists
 - After you know you can put the airplane safely on the ground somewhere, the next step to increase the likelihood of a safe outcome is to try and restart the

engine. The emergency checklists should be memorized in a flow pattern. If altitude permits, it is also recommended to verify no line items were missed with the physical checklist.

- D - Declare
 - Aviate, Navigate, Communicate. If safe to do so, declare an emergency on 121.5 and squawk 7700.
 - "mayday mayday mayday (tail number) (location) (emergency) (souls on board) mayday mayday mayday".
 - You can press and hold the frequency toggle soft button to automatically put 121.5 in the active frequency box in COM 1.

Emergency Approach and Landing

- This is performed to simulate a situation where power is lost in flight.
- Ensure that the student is familiar with the procedures and has the checklists memorized!
- Quiz the student on the procedures and address any questions they might have.

Emergency Descent

- This is performed to simulate a situation in which the airplane needs to descend in the most expeditious means possible.
- Smoke in the cockpit, electrical fire, hypoxic passenger, depressurization, general emergency situations.
- Quiz the student on the procedures for the maneuver and address any questions they might have.

Normal Traffic Pattern and Powered Approach

- Ensure that the student knows how to conduct a proper traffic pattern by quizzing them on:
 - Entry procedures.
 - Airspeeds in the pattern.
 - Flap usage in the pattern.
 - Power settings.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.

- NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
- Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
- FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all the checklists.
- Student should be able to perform all briefings.

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Operation of the Communications and Navigation Units

- Ensure that the student understands how to properly use COMS and NAV units.

Radio Communication Procedures and ATC Phraseology

- Students should understand and demonstrate:
 - Proper radio setup and frequency management
 - Initial call-up format (who you are calling, who you are, where you are, what you want)
 - Accurate readback of clearances and instructions (runway assignments, hold short instructions, altitudes, headings, squawk codes)
 - Use of standard terminology (e.g., "affirmative," "negative," "standby," "unable")
 - Avoid nonstandard or casual language
 - Maintaining situational awareness while communicating
- The student should be able to conduct radio calls in the practice area.
- The instructor should assist as needed.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Normal and Crosswind Takeoff

- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.

Review Previous Exercises as Required

- Review any areas the student might need help with by utilizing the pertinent lesson plans.

Emergency Operations

- Engine Failure in Flight (Emphasis on Pattern)
 - Student says, student does.
 - Verify correct procedure.
 - Points of Emphasis.
 - Trimming for V_{glide} .
 - Wire strike avoidance.
 - Choice of landing point.
 - Coach the student through a proper flow while going through the checklist (left to right!).
- Emergency Descent
 - Student says, student does.
 - Verify correct procedure.
 - Points of Emphasis.
 - Trim.
 - Area clearance.
 - Ensure the student does not exceed V_{FE} .

Ground Reference Maneuvers

- Student says, student does.
 - Verify correct procedure.
- Points of Emphasis.
 - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
 - Bank variation.
 - Choosing an adequate and safe visual reference point.
 - Provide feedback as needed throughout the maneuver.

Normal and Crosswind Landing

- The student should conduct the landing.
- Provide guidance during the round and flare.
- Provide guidance on proper crosswind correction.
- Emphasize that this must be maintained even after touchdown.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique
- Instructor critique.
- Questions from the student.

- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 119.
- Assign homework:
 - Review the emergency checklists.
 - Review ground reference maneuvers.

Completion Standards

This lesson is complete when the applicant can:

1. Demonstrate proper sequential checklist procedures for normal procedures without instructor guidance.
2. Conduct emergency procedures with little instructor guidance.
3. Correctly perform previously learned maneuvers with minimal instructor guidance.
4. Conduct takeoffs and rejected takeoffs with minimal instructor guidance.
5. Conduct landings with instructor guidance.
6. During ground reference maneuvers, with minimal instructor guidance:
 - a. Fly a predetermined ground track using specific ground references.
 - b. Understand the effects of wind and correct for wind drift.
 - c. Maintain altitude ± 200 ft, and airspeed ± 15 knots, and a maximum bank of 45°
7. Make radio calls with little instructor guidance.